

NETWORKS AND COMPLEXITY

Solution 17-5

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 17.5: Manifold approximation [Lvl. 2]

Find a second order (quadratic) approximation to the invariant manifolds of the steady state at $(0, 0)$ in the following system:

$$\begin{aligned}\dot{x} &= 2x + 4y - x^2 \\ \dot{y} &= x - y + xy\end{aligned}$$

Solution

We choose the Ansatz

$$x = ay + by^2 + O(y^3) \quad (1)$$

Differentiating the Ansatz yields

$$\dot{x} = (a + 2by)\dot{y} + O(y^3) \quad (2)$$

Now we substitute the equations of motion from the exercise

$$2x + 4y - x^2 = (a + 2by)(x - y + xy) + O(y^3) \quad (3)$$

We can now substitute the Ansatz itself.

$$2(ay + by^2) + 4y - (ay + by^2)^2 = (a + 2by)((ay + by^2) - y + (ay + by^2)y) + O(y^3) \quad (4)$$

We need to multiply all these terms out. However, everything that contains y^3 can be immediately be absorbed into the $O(y^3)$ terms.

$$2ay + 2by^2 + 4y - a^2y^2 = (a + 2by)(ay + by^2 - y + ay^2) + O(y^3) \quad (5)$$

$$2ay + 2by^2 + 4y - a^2y^2 = a^2y + aby^2 - ay + a^2y^2 + 2aby^2 - 2by^2 + O(y^3) \quad (6)$$

$$(3a + 4 - a^2)y = (2a^2 + 3ab - 4b)y^2 + O(y^3) \quad (7)$$

Since the equation must be true for every y (all the way along the manifold) the coefficients of the y and y^2 terms must be zero. We first consider the coefficient of y

$$a^2 - 3a - 4 = 0 \quad (8)$$

$$(a - 4)(a + 1) = 0 \quad (9)$$

So a is -1 or 4 which one should we chose? Actually recall from the previous exercise (or compute it now) that the stable eigenvector was $(-1, 1)^T$. The manifold must be tangential to it and hence we need a function that has slope -1 in $x = 0$ to point in the direction of the eigenvector. So it has to have the form $x = -y + O(y^2)$. This means we could have inferred

$a = -1$ just from the previously computed eigenvector, but it's a good check to see it appear also in this way. So now we know for sure $a = -1$. Let's substitute it in the coefficient of y^2 ,

$$0 = 2a^2 + 3ab - 4b \quad (10)$$

$$0 = 2 - 3b - 4b \quad (11)$$

$$0 = 2 - 7b \quad (12)$$

$$b = \frac{2}{7} \quad (13)$$

In summary our approximation to the manifold is

$$x = -y + \frac{2}{7}y^2 \quad (14)$$

If you are wondering if this is really the stable manifold and not the unstable one, you can substitute this into the equation for $\dot{y}$:

$$\dot{y} = x - y + xy = -2y + O(y^2) \quad (15)$$

Hence close to the steady state at zero the equation behaves like

$$\dot{y} = -2y \quad (16)$$

which shows that it is stable.

The unstable manifold can be computed in the same way. The result is

$$x = 4y - 4y^2 + O(y^3) \quad (17)$$